



Computer Graphics|CS2206

Code Queens



Group 4



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Task Table

	Developed Scene 1 + prepared the project report.
	Developed Scene 3 + handled the final code integration and modifications.
	Developed Scene 4 + prepared the presentation.
Huda Al-Halaf	Developed Scene 2.



The Introduction

This project presents the design and implementation of an interactive graphical application using C++ and OpenGL. The project is structured as a simple story-based system that consists of multiple scenes, where each scene represents a different stage within the application.

The main objective of this project is to apply fundamental concepts of computer graphics by creating interactive visual environments. It also aims to enhance understanding of scene management and user interaction through keyboard input.

The system allows the user to navigate between scenes and interact with various graphical elements. Each scene includes visual components such as shapes, colors, and text, which are combined to create a clear and organized graphical environment.

Overall, the project demonstrates how multiple graphical components can be integrated into a single cohesive application to provide a basic interactive experience.



Methodology

Scene 1: The Classroom

Scene 1 represents the classroom environment and serves as the project's starting point. This scene introduces the user to the graphical interface using simple visual elements. The user can navigate the scene using the keyboard keys "A," "D," "W," and "S" to move in different directions. This scene allows the user to progress to the next stage of the .project

Scene 2: The Library

Scene 2 represents the library environment. It includes visual elements such as books, shelves, and furniture to simulate a study space. This scene enhances the visual experience and completes the .project's development



Scene 3: The Puzzle Room

Scene 3 is an interactive puzzle room where the user can move objects using the keyboard. The goal is to arrange or place these objects in the correct location, adding an interactive challenge to the .project

Scene 4: The Digital World

Scene 4 represents a digital world environment with a nighttime theme. It includes elements such as buildings, windows, and the moon. The user can interact with this scene by controlling the window lighting and changing the moon's shape, providing .an interactive conclusion to the project



Libraries and Tools Used

In this project, several libraries were used to support graphics rendering, user interaction, and sound effects.

The **OpenGL** and **GLUT** libraries were used to create and display graphical elements such as shapes, colors, and text. These libraries provide the core functionality for rendering the scenes.

The **GLEW** library was included to ensure compatibility with modern OpenGL features. Conditional compilation was also used to support different operating systems.

Additionally, the Windows multimedia library (**winmm**) was used to handle sound effects within the project, allowing audio to be played during interaction.

Other standard libraries such as **cmath**, **cctype**, and **iostream** were used to support mathematical operations, character handling, and general program functionality.



Result

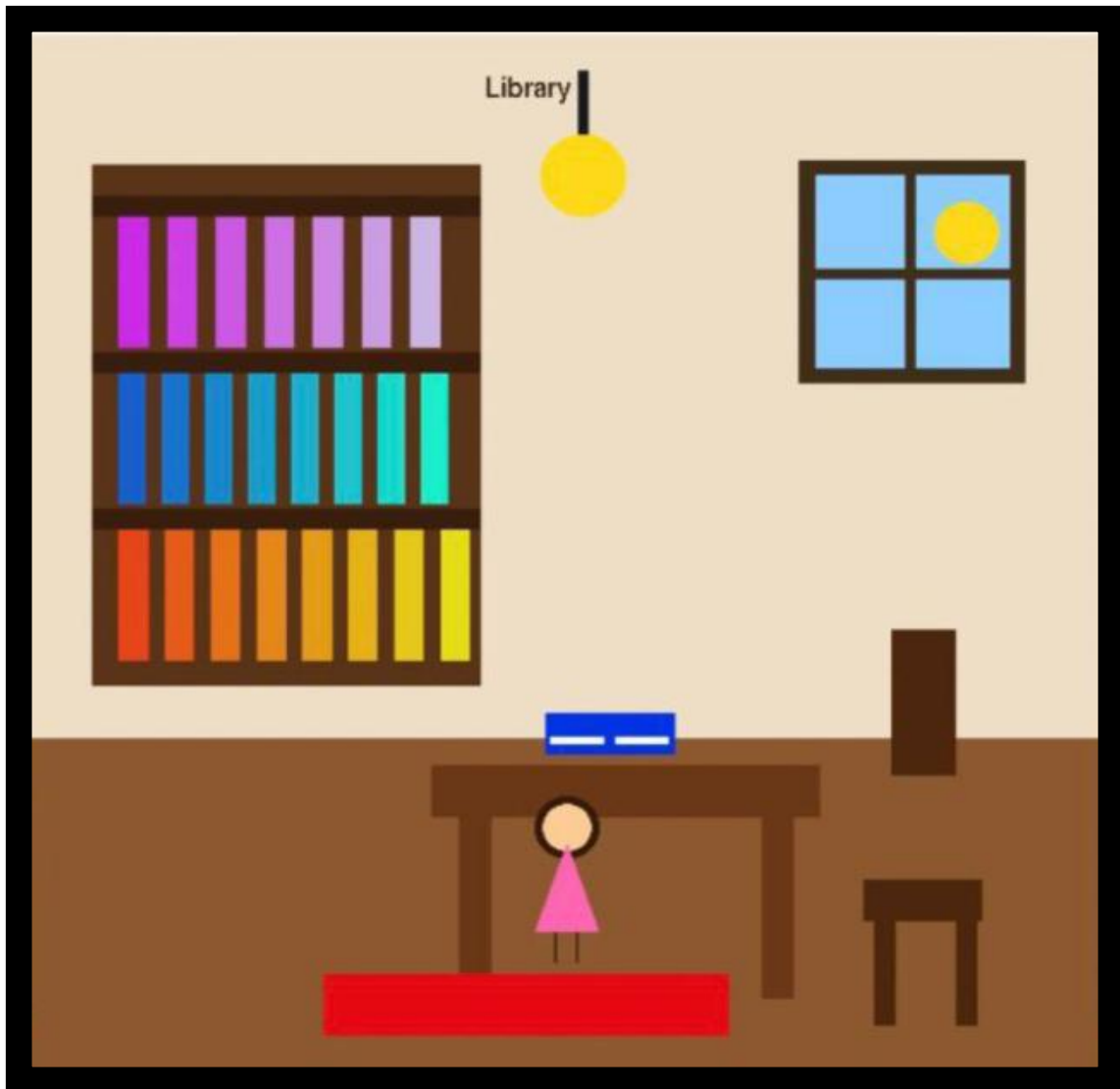
Scene 1: Class room



The keys "A", "D", "W", and "S" are used to move the character left, right, up, and down within the classroom environment.



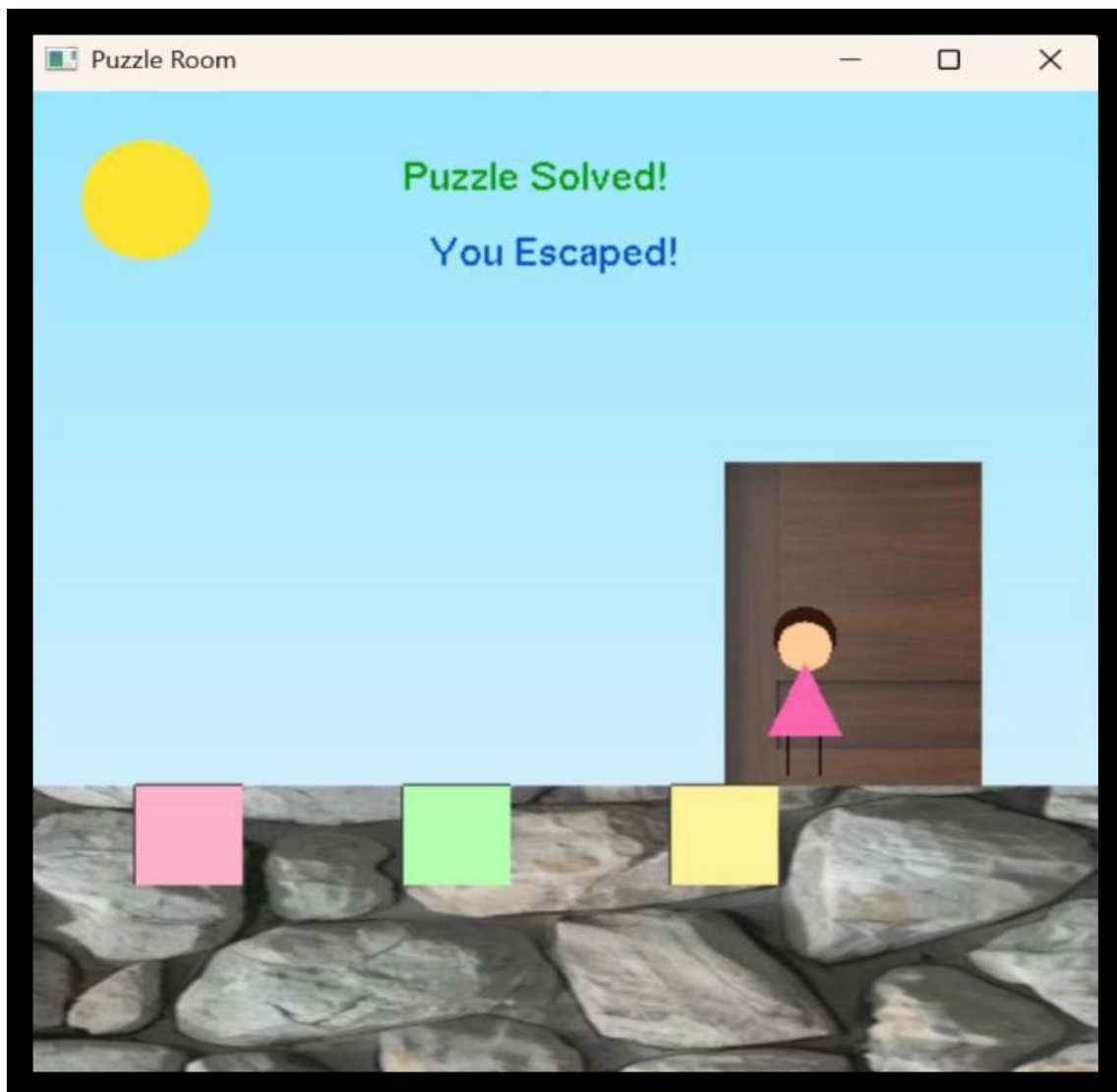
Scane 2 : Library



The keys "A", "D", "W", and "S" are used for navigation. In addition, the key "I" is used to control the lamp, while the keys "O" and "C" are used to open and close the book.



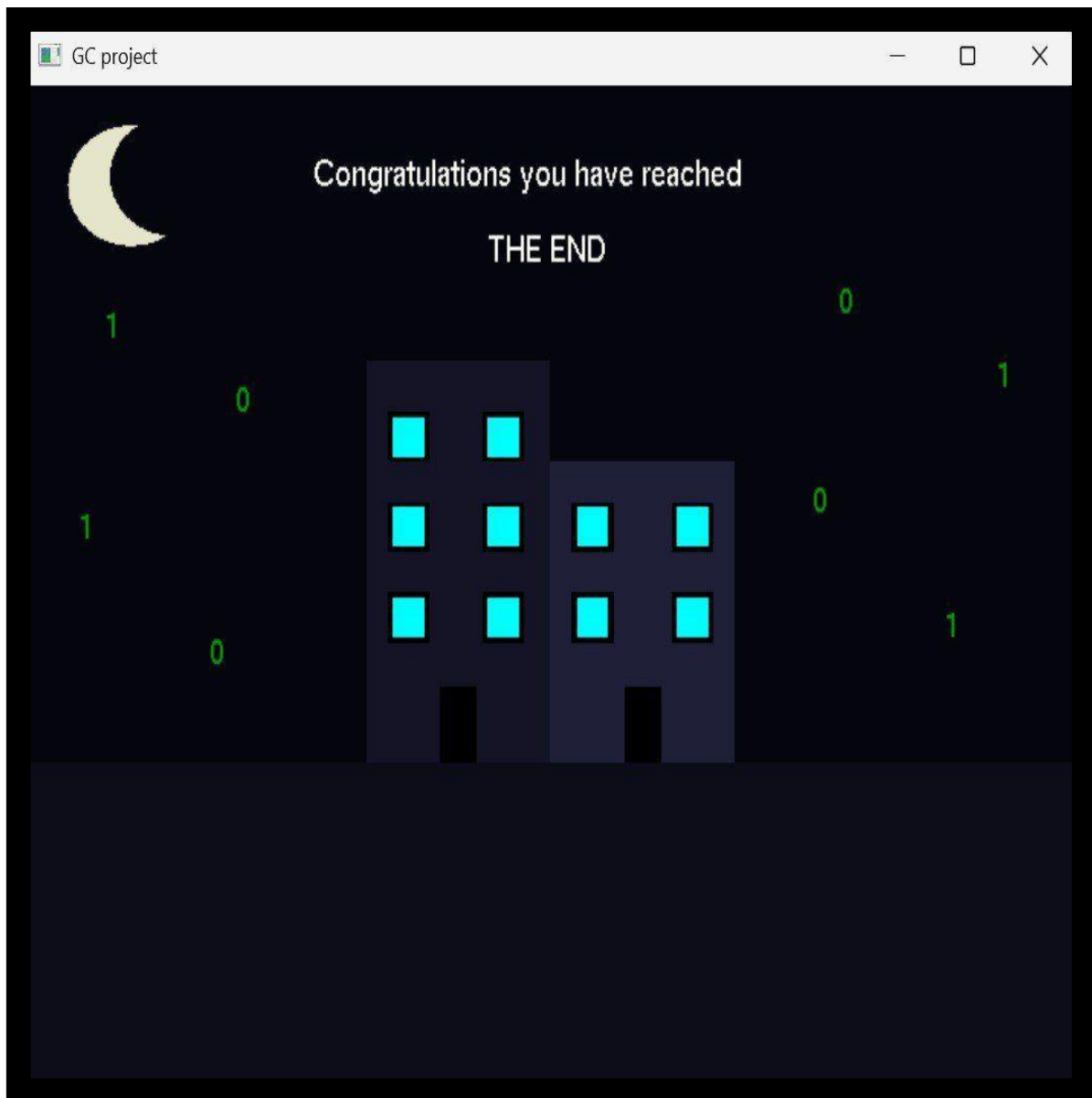
Scane 3: Puzzle room



The keys "1", "2", and "3" are used to control and position the cubes, allowing them to move to their correct places



Scane 4: Digital world



The key "L" is used to control the building lights, while the key "M" is used to change the shape of the moon.



Conclusion and Future Work

In conclusion, this project successfully developed an interactive graphical application using C++ and OpenGL, combining multiple scenes into one integrated system with user interaction. It demonstrated important concepts such as scene management, graphical rendering, and handling .user input

For future work, the project can be improved by enhancing the visual quality of the scenes through better design and smoother animations. Sound effects can also be expanded by adding background audio and more interactive feedback. In addition, the interaction can be developed further by .introducing more complex puzzles and challenges

The project can also be extended by adding a clearer storyline that connects all scenes, as well as creating additional levels to improve the overall user experience and make the application more engaging.



References

- OpenGL Applicatio
- Computer Graphics Book